CLASS 10 — SCIENCE · FORMULA SHEET

Chapter 9: **Light - Reflection and Refraction** | Quick revision, one page

Reflection

- Laws: angle of incidence = angle of reflection; incident ray, reflected ray and normal lie in the same plane.
- Plane mirror: image virtual, erect, same size, laterally inverted, as far behind as the object is in front.
- Spherical mirrors: $f = \frac{R}{2}$; principal focus F; concave converges, convex diverges.
- Mirror formula: $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$. Magnification: $m = \frac{h'}{h} = -\frac{v}{u}$.

Sign convention (Cartesian)

- Pole is origin; principal axis is x-axis; object on the left. Distances measured against the incident light are negative.
- Concave mirror: $f < 0$; convex mirror: $f > 0$. Heights upwards positive, downwards negative ($m < 0$ real, inverted; $m > 0$ virtual, erect).

Images by concave mirror

- Object at infinity: image at F, real, point. Beyond C: between F and C, real, inverted, diminished. At C: at C, real, inverted, same size.
- Between C and F: beyond C, real, inverted, enlarged. At F: at infinity, enlarged. Between P and F: behind mirror, virtual, erect, enlarged.

Refraction

- Laws: incident ray, refracted ray, normal in one plane; Snell's law $\frac{\sin i}{\sin r} = n$ (constant).
- Refractive index $n = \frac{c}{v}$ (c = 3×10^8 m/s). Rarer to denser: bends towards normal; denser to rarer: away from normal.
- Rectangular glass slab: emergent ray parallel to incident ray (lateral shift).

Lenses

- Lens formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$; magnification $m = \frac{v}{u} = \frac{h'}{h}$.
- Power: $P = \frac{1}{f}$ (f in metres); unit dioptre (D); convex +, concave $-$. For lenses in contact, $P = P_1 + P_2$.
- Convex lens: $f > 0$; real inverted image when $u > f$; virtual erect enlarged when object between O and F. Concave lens: always virtual, erect, diminished.